ProofMint

Verifiable AI Agent Certificates on Hedera

HTS (NFTs) | HCS (Audit Trail) | Mirror Node (Verification)

Hedera Hello Future Apex 2026 | AI & Agents Track

Diven Rastdus | Solo Builder

THE PROBLEM

AI agents are black boxes. Nobody can verify what they did.

No Proof of Work

An AI agent says it reviewed your code, analyzed your data, or passed your compliance check. But where's the evidence? Logs can be edited. Databases can be tampered with.

No Accountability

When AI makes decisions that affect real systems, there's no immutable record of what was analyzed, what was found, and what was decided. Trust is assumed, not verified.

> *"Did the AI actually review this, or did someone just click approve?"*

THE SOLUTION

Mint an NFT certificate for every AI task

ProofMint creates a permanent, verifiable, on-chain record every time an AI agent completes work.

AI Reviews Code → Publish to HCS → Mint NFT Certificate → Verify via Mirror Node

Immutable Audit

Full task details (hashes, verdict, issues) published to HCS. Cannot be altered after the fact.

NFT Certificate

Compact NFT minted via HTS referencing the HCS message. Transferable, permanent proof.

Independent Verify

Anyone can query Mirror Node to cross-reference NFT metadata against HCS audit trail.

HEDERA INTEGRATION

Three services, structurally essential

SERVICE	ROLE	WHAT HAPPENS
HTS (Token Service)	Certificate Registry	NFT collection created; each task mints a new serial with compact metadata
HCS (Consensus Service)	Audit Trail	Full task details (input hash, output hash, verdict) published as ordered messages
Mirror Node	Verification Layer	REST API queries fetch NFT metadata + HCS messages for independent hash comparison

Remove any one service and the system breaks. Without HTS: no transferable certificates. Without HCS: no audit trail to reference. Without Mirror Node: no independent verification.

ARCHITECTURE

Tech Stack & On-Chain Proof

Stack

- **hiero-sdk-python** - HTS + HCS transactions
- **FastAPI** - Python backend + REST API
- **Next.js 15** - Certificate dashboard
- **SQLite** - Local state (verified against chain)
- **AI Code Reviewer** - Rule-based + Gemini

Confirmed On-Chain (Testnet)

- NFT Collection: **0.0.8114539**
- HCS Topic: **0.0.8114364**
- 6 NFT serials minted
- 10+ HCS audit messages
- All verifiable on HashScan

TokenCreateTransaction | TokenMintTransaction | TopicCreateTransaction | TopicMessageSubmitTransaction | Mirror Node REST

WHY HEDERA

The only network where this is economically viable

$0.001

Per NFT mint on Hedera. Ethereum: $2-50+ per mint.

Native HCS

Ordered message logging without smart contracts. No Solidity needed.

3-5s

Finality. Real-time certificate minting after each AI task.

Per-task NFT minting is a use case that **only works on Hedera**. On any EVM chain, minting an NFT for every AI code review would cost more than the review itself. Hedera's fixed, predictable fees make high-frequency certification practical.

JUDGING CRITERIA

How ProofMint scores

CRITERIA	PROOFMINT
Innovation	First system to mint per-task NFT certificates for AI agents using HTS + HCS combined
Technical Execution	3 Hedera services deeply integrated (HTS, HCS, Mirror Node). All transactions confirmed on testnet.
User Experience	One API call to submit code → certificate appears in dashboard with HashScan links
Hedera Usage	HTS (NFT collection + minting), HCS (audit trail), Mirror Node (verification). Remove any = broken.
Practicality	Solves real trust gap in AI agent outputs. Applicable to code review, compliance, auditing.
Scalability	At $0.001/mint, scales to millions of certificates. Hedera throughput: 10K+ TPS.
Presentation	Full demo video, pitch deck, live dashboard, on-chain proof on HashScan.

FUTURE ROADMAP

Where ProofMint goes next

Phase 1

Multi-agent support: any AI agent (not just code review) can mint certificates

Phase 2

Certificate marketplace: companies verify and trade AI work credentials

Phase 3

Mainnet launch with SDK for developers to integrate ProofMint into their AI pipelines

Try it yourself

github.com/astraedus/proofmint

hashscan.io/testnet/token/0.0.8114539

hashscan.io/testnet/topic/0.0.8114364

Built by Diven Rastdus | theagentthatcould@gmail.com